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MILITARY SPECIFICATION

BENDING MACHINES, ROTARY HEAD, TUBING, HYDRAULICALLY OPERATED, NUMERICALLY CONTROLLED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hydraulically operated, tube bending machines of the rotary head type, utilizing a numerically controlled positioning system with equipment and accessories specified herein.

1.2 Classification. Bending machines shall be of the following classes and sizes listed in table I. The class and size to be furnished shall be as specified (see 6.2.1 and table I).

Class A - Bending machines with counter-clockwise bending motion (left hand).
Class B - Bending machines with clockwise bending motion (right hand).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristic Measuring of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

CFR 29, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.15 - Pipe, Tube, and Shape Bending Machines Safety Requirements for the Construction, Care, and Use.

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(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One inch Perforated Paper Tape.

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

RS-281 - Electrical and Construction Standards for Numerical Machine Control.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006-1899.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other information services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The bending machine shall be new and one of the manufacturer's current models capable of making horizontal, rotary bends of tight radii in standard and thin wall tubing in both ferrous and non-ferrous materials. Bending shall be accomplished by wrapping the workpiece around a bending form or die. Design shall include provisions for mounting tooling consisting of bending forms, clamping dies, pressure dies, wiper dies, and mandrels. Mounting shall permit complete changeover from one bending pattern and tube diameter to another. The bending machine shall be designed for use in conjunction with the numerical control system specified herein. The system shall be designed to automatically form multiple bends with each bend having different distance between bends (DBB), rotation or plane of bends (POB), and degree of bends (DOB). All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

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3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the bending machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of CFR 29, Chapter XVII, Part 1910 and ANSI B11.15 that are applicable to the machine itself. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. When specified (see 6.2.1), a centralized automatic lubrication system shall be provided.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

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3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Dials. The diameter of the dial shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The bending machine shall consist of not less than a base, nose assembly, spindle, arms, pressure die assembly, clamp die assembly, mandrel lubricator, mandrel extractor, indexing carriage, wiper die assembly, numerical control system, tooling, hydraulic system, and electrical system.

3.4.1 Base. The base structure shall be either cast iron, cast steel, or a fabricated steel weldment of the mass, weight, and rigidity to support all components of the machine while operating at the maximum rated capacity without permanent distortion. The base and oil reservoir shall be integral or the oil reservoir shall be mounted inside the base. The base shall have adaquate means to permit leveling and fastening the machine to the floor or foundation.

3.4.2 Nose assembly. The nose assembly shall be a steel casting or weldment attached to the base. The nose shall be bored to accomodate precision antifriction bearings or otherwise arranged to accept a cartridge type

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spindle housing which has been bored to accept precision antifriction bearings.

3.4.3 Spindle. The spindle shall be of precision ground alloy steel, sufficient in size and strength to withstand the maximum forces applied when the machine is operating at the maximum rated capacity. The spindle shall rotate on precision antifriction bearings mounted in the nose and shall have provisions for mounting and locking in place the drive spindle or sprocket and rotating bending arm.

3.4.4 Stationary and bending arms. The stationary and bending arms shall be steel castings or steel plate, or a combination thereof, sufficient in strength to withstand the maximum forces applied when the machine is operating at the maximum rated capacity. The arms shall be accurately machined and fitted with slides mounted on replaceable wear strips or an adjustable gib shall be provided for lateral wear adjustment. The die mounting boss of the bending arm shall be fitted with keys and a locating ring for accurately positioning the bending dies.

3.4.5 Pressure die assembly. The pressure die assembly shall be mounted on the stationary arm and shall operate by means of a direct acting hydraulic cylinder or thru a hydraulic cylinder actuated, self-locking linkage assembly. The cylinder shall be powered either from an independent hydraulic power source or from the main hydraulic system, provided that the pressure die cylinder is isolated from any fluctuation of the pressure in the main hydraulic system. A gauge for reading the pressure on the pressure die and a means for adjusting the pressure shall be provided. The pressure die assembly ways shall be faced with replaceable wear strips or shall incorporate a gib to permit adjustment and a seal to prevent entry of foreign matter. Provisions for lateral adjustment and compensation for variances in tubing sizes to suit dies for all radii within the capacity of the machine shall be included. The pressure die assembly master bar shall incorporate vertical adjustment for the pressure dies and a provision to allow for automatic compensation for the normal diameter variations that occur between workpieces of the same nominal size. The pressure die master bar shall automatically return after the pressure die has been opened. The pressure die master bar may have a hydraulic booster cylinder or pressure die assist if necessary to provide the performance as specified herein.

3.4.6 Clamp die assembly. The clamp die assembly shall be mounted either on the bending form or a swing arm. When one specific type clamp configuration is desired it shall be as specified (see 6.2.1) The clamp assembly shall operate by means of a hydraulic cylinder actuating linkage or a direct acting hydraulic cylinder. Either means of operation shall provide for clamping pressure up to the maximum force needed for clamping all material within the capacity of the machine. The clamp die assembly shall incorporate means for wear compensation and the ways shall be faced with replaceable wear strips. The die holder shall provide mounting and adjustment of the dies to permit distribution of pressure over the entire clamp area.

3.4.7 Mandrel lubricator. When specified (see 6.2.1), a mandrel lubricator designed to use any type of drawing compound currently in use in the areospace

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industry shall be supplied. The lubricator shall be operable from the console and from programming as an auxiliary function.

3.4.8 Mandrel extractor unit. The mandrel loading and extracting unit shall include a mandrel rod, supports, hydraulic cylinder, powered adjustment, and any other component or feature necessary to permit location of the mandrel at a predetermined position and withdraw it automatically after completion of a bend. Provision shall be made to prevent accidental rotation of the mandrel during operation.

3.4.9 Indexing carriage. A power operated indexing carriage shall be furnished to facilitate loading the tube over the mandrel, moving the tube forward to the correct distance between bends, and rotating the tube to change plane of bend. Machines supplied with an indexing carriage shall have a drop away clamp or other means to prevent interference with the bent pipe during the retraction cycle of the machine.

3.4.10 Wiper die assembly. A wiper die assembly to mount wiper dies in proper bending position shall be provided. Provisions shall be made for vertical, longitudinal, lateral, and radial positioning of the wiper die to the bend die, with sufficient movement to rigidly mount wiper dies for all radii within the machine capacity.

3.4.11 Computer numerical control (CNC). The tubing bender shall be equipped with a fully automatic, solid state, softwired, integrated circuit type numerical control capable of providing automatic control of machine functions and operating modes for the distance between bends, plane of bend, and degree of bend. The control shall automatically compensate for springback by providing the proper overbend to allow for tube springback of the programmed bend angle up to and including 180 degrees. Other functions including operation of the clamp die, pressure die, collet, mandrel, and mandrel lubrication shall be automatically sequenced by the numerical control. The mechanical and electrical design of the control shall comply with EIA RS-281 construction standards. Unless otherwise specified (see 6.2.1), the control shall be provided for connection of a 115-volt, alternating-current (AC), 60-Hz, single phase power source. Line voltage fluctuation of ± 10 percent from the normal operating voltage shall not adversly affect the operation of the CNC.

3.4.11.1 Control system modes of operation. Modes of operation shall include automatic, step, and manual data input (MDI). In the automatic mode all machine functions for a complete tube bending program shall be automatically executed by the CNC. In the step mode a specific sequence of the program shall be selected and executed independently by the operator through the MDI keypad. In the manual mode each machine movement shall be directed by the operator through MDI.

3.4.11.2 Control memory. Part program memory resident in the control shall have a capacity of not less than 7 programs consisting of at least eight bends each. The control memory shall be capable of storing part program data consisting of the numeric data required for calculations (tool dimensions, length of end of tube, and springback compensation factors) and numeric designations representing the tube configurations (DBB, POB, and DOB). Part

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program data shall be entered into the memory by MDI, an auxiliary storage media, or a tube measuring system. Controls having volatile part program memory shall have a battery backup capable of retaining part programs in memory for not less than 7 days.

3.4.11.3 Permanent memory. A magnetic cassette tape, floppy disk, or one inch perforated tape system shall be provided for additional memory storage. The specific type memory storage desired shall be as specified (see 6.2.1). When one inch punched tape data input is required it shall be in compliance with EIA RS-227. The control shall be equipped with an interface complying with EIA RS-232.

3.4.11.4 Tape preparation unit. When specified (6.2.1), a tape preparation unit capable of producing an original or modified tape from the control memory unit shall be furnished.

3.4.11.5 Part program edit and display. A program edit capability shall be provided to allow insertion, deletion, and modification of any part program in the part program memory. Data related to part program edit shall be entered by the manual data input provided on the control panel. Edited program data entered into the program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled. The system supplied shall be capable of displaying in easily understood form, all operating data, actual and command, from memory and active data.

3.4.11.6 Tube measuring data system. When specified (see 6.2.1), a tube measuring data system capable of interpreting data on bent tube shapes from all conventional sources including X, Y, and Z coordinates, blueprints, master tube shapes, templates, and inspection fixtures shall be provided. The system shall consist of an articulating measuring arm with a minimum of three axes, a mini computer, a cathode ray tube (CRT) display, and a MDI keypad. The tube measuring data system shall be capable of reversing a bend program end for end to overcome possible machine interference, calculating raw material length, and compensation for elongation and springback. An interface complying with EIA RS-232 to the control shall be provided. Tube capacity including size of support table, and range of articulating arm shall be as specified and fully described (see 6.2.1).

3.4.12 Tooling. When specified (see 6.2.1), tooling consisting of bending dies, clamp dies, pressure dies, wiper dies, and mandrels shall be furnished. When ordering pressure dies and wiper dies the outside diameter, wall thickness, material and alloy of the tubing to be bent shall be specified. Orders for bending dies and clamp dies shall specify the outside diameter, wall thickness, material and alloy of the tubing to be formed, the centerline bend radius, and maximum bend angle required. Orders for mandrels shall specify outside diameter, wall thickness, material and alloy of the tubing with which they will be used, the type mandrel, and the minimum centerline bend radius. Tooling ordered shall include all associated components and hardware to assure compatibility with the machine specified. Tooling shall have the hardness and surface finish compatible with the type tubing specified to prevent galling and other malfunctions during the bending process.

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3.4.13 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level. Unless otherwise specified (see 6.2.1), the hydraulic system shall conform to JIC H-1.

3.4.14 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating. Unless otherwise specified (see 6.2.1), all AC controls and motors shall operate on 60 Hz electrical power.

3.4.14.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure.

3.4.14.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.14.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. Bending machines covered by this specification shall be capable of bending ferrous and non-ferrous standard and thin wall tubing of the sizes and capacities and to the degrees specified in table I. Complete tooling changeover from one bending pattern and tube diameter to another shall be accomplished within 30 minutes. The 30 minute time span shall include changing and adjusting the tooling and entering new part program data into the control system memory. The bending machine shall perform to the accuracy requirements of table II when tested in accordance with 4.7.4.1 and 4.7.4.2.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table III.

3.8 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

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TABLE I. Sizes and capacities, class A and B.

Capacities		Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8
Bending ranges: Tubing sizes (outside diameter x wall thickness, inches)									
Aluminum	min.	0.187x0.022	0.375x0.022	0.50x0.028	0.750x0.028	1.0x0.125	1.5x0.200	2.0x0.218	2.5x0.203
	max.	0.750x0.125	1.0x0.179	2.0x0.218	3.0x0.188	4.0x0.375	6.0x0.432	8.0x0.50	10.0x0.593
Mild steel	min.	0.187x0.022	0.375x0.028	0.50x0.035	0.750x0.109	1.0x0.118	1.5x0.145	2.0x0.154	2.5x0.120
	max.	0.750x0.079	1.0x0.133	2.0x0.154	3.0x0.109	4.0x0.250	6.0x0.280	8.0x0.322	10.0x0.365
Stainless steel	min.	0.187x0.028	0.375x0.028	0.50x0.035	0.750x0.035	1.0x0.095	1.5x0.109	2.0x0.109	2.5x0.083
	max.	0.750x0.045	1.0x0.050	2.0x0.095	3.0x0.095	4.0x0.125	6.0x0.134	8.0x0.148	10.0x0.250
Bend radius to centerline (inches)		2D to 5	2D to 6	2D to 11	2D to 11	2D to 18	2D to 20	2D to 32	2D to 38
Tube length over mandrel (inches)		96	96	120	120	162	162	170	198
Bend angle max. (degrees)		180	180	180	180	180	180	180	180
Operating speeds: Distance between bends: Carriage travel (inches per minute)		800	750	450	450	400	400	250	139
Resolution (inches)		±0.01	±0.01	±0.01	±0.01	±0.01	±0.01	±0.01	±0.01
Plane of bend: Collet rotation (rpm)		30	30	30	22	8	8	3	3
Resolution (degrees)		±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1
Degree of bend: Bend arm rotation (rpm)		12	12	8.5	10	3	3	1.3	.7
Resolution (degrees)		±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1	±0.1

NOTE: Requirements are not less than those shown. Where a range is shown, requirements are from the smaller figure or less to the larger figure or more.

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TABLE II. Performance accuracies.

Bending angles (DOB)	± 1/4 degree
Plane of bend (POB)	± 1/4 degree
Flattening (not more than)	4%
Wall thinning (not more than)	35%

TABLE III. Alignment tolerances classes A and B.

Die boss surface parallel with stationary and swinging arm way and slides	0.015 inch TIR
Runout of spindle and die boss in 180 degrees	0.005 inch TIR
Pressure die master bar to move parallel to die boss	± 0.005 inch TIR over 12 inch travel
Pressure die mounting surface perpendicular to die boss surface	90 degree ± 10 min.
Wiper die holder mounting surface parallel to die boss (over full length)	± 0.005 inch
Accuracy of machine members:	
Carriage travel	± 0.01 inch
Collet rotation	± 0.1 degree
bending arm rotation	± 0.1 degree

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3.9 Optional equipment. When required (see 6.2.1) the following equipment shall be furnished for the tubing size specified:

a. Bend dies
b. Clamp dies
c. Pressure dies
d. Wiper dies
e. Pressure die boost assembly
f. Mandrels
g. Mandrel extractor

3.10 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.11 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.11.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

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3.13 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.14 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance

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inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. Functions of the machine and control unit shall be tested in accordance with the machine tool builder's standard procedure for warm-up and run-off tests. The run-off tests shall include operating the machine in each of its modes of operation to determine proper response of the machine to control commands. Upon completion of the warm-up and run-off tests, the machine shall be cycled continuously for a period of 4 hours with out a malfunction. Should a failure occur, the cycle shall be restarted after the correction and run for a full 4 hour period.

4.7.2 Alignment tolerance test. The machine shall be tested in accordance with the machine tool builder's standard test methods to determine that the machine conforms to the alignment tolerances of table III.

4.7.3 Deflection tests. Using a set-up for maximum diameter tubing and wall thickness and utilizing required clamp and die pressure, the machine shall be measured for deflection using the tests specified in 4.7.3.1, 4.7.3.2 and 4.7.3.3.

4.7.3.1 Deflection between bending die and clamping die. The deflection between the bending die and clamping die shall be measured at the centerline of the clamping pressure. The deflection shall not exceed ±0.015 inch.

4.7.3.2 Spindle deflection. The spindle deflection shall be measured by taking a dial indicator reading at the tangent of the bend form. The bend form shall be rotated 90 degrees and the dial indicator shall again be read at the same point on the bend form. Spindle deflection shall not exceed ± 0.010 inch.

4.7.3.3 Wiper die deflection. The wiper die deflection shall be measured at the center line of the pressure die cylinder. Wiper die deflection shall not exceed ± 0.010 inch on size 1 machines, ± 0.015 inch on sizes 2 and 3 machines, ± 0.020 inch on sizes 4, 5, and 6 machines, and ± 0.030 inch on sizes 7 and 8 machines.

4.7.4 Performance tests.

4.7.4.1 Automatic Mode. The machine and control shall be set up for automatic mode of operation. The configuration of the test parts and bending program shall be the supplier's choice, designed to most effectively utilize the capabilities of the size machine and control being tested. Not more than two sample bends shall be allowed. Two identical tubes shall be processed without operator assistance except for loading and unloading. Not less than five bends shall be made in each tube with each bend having a different distance between bends, plane of bend, and degree of bend. The completed test parts shall be measured for compliance with the accuracies of table II. The 30

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minute tooling changeover shall be tested to ensure conformance with the requirements of 3.6

4.7.4.2 Step mode. Utilizing the same configuration of test parts, bending program, type and size of tubing as used in 4.7.4.1, two identical tubes shall be processed with the machine and control in the step mode of operation. The completed test parts shall be measured for compliance with the accuracies of table II.

4.7.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9. The test shall be accomplished in accordance with MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The bending machines covered by this specification are intended for use in manufacturing and repair facilities for bending standard and thin wall tubing where precision tolerances and exact repeatability are required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2 and table I).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US or SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

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h. Centralized automatic lubrication system, if required (see 3.2.9).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

k. If one specific type clamp configuration is required, specify type (see 3.4.6).

l. Mandrel lubricator, if required (see 3.4.7).

m. Numerical control input power source, if different (see 3.4.11).

n. Permanent memory storage,specify type (see 3.4.11.3).

o. Tape prepartion unit, if required (see 3.4.11.4).

p. Tube measuring data system, if required (see 3.4.11.6).

q. Specify and fully describe, capacity of tube measuring system (see 3.4.11.6).

r. Tooling, if required, fully describe (see 3.4.12).

s. Hydraulic system, if different (see 3.4.13).

t. Electric system and input power, if different (see 3.4.14).

u. Motor and control frequency, if different (see 3.4.14).

v. Motor enclosure , if different (see 3.4.14.1).

w. Size and capacity, if different (see 3.5).

x. Additional performance features, if required (see 3.6.1).

y. Standard equipment, if different (see 3.8).

z. Specify and fully describe optional equipment, and tubing size if required (see 3.9).

aa. Specify applicable repair parts, if required (see 3.10).

bb. Hour meter, if different (see 3.12).

cc. First article inspection, if different (see 4.4).

dd. Quality conformance inspection, if different (see 4.5).

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ee. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

a. Sizes five, six, seven, and eight were added to MIL-B-80161C.

6.5 Subject term (key word) listing.

Bender,tube
Computer numerical control
Dies, bending
Hydraulic
Mandrel
Metal, fabricating
Rotary bender

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Air Force - 99

Review activity:
Air Force - 84

User activities:
Army - AV
Navy - MC, SH

Preparing activity:
DLA - IP

(Project 3441-0113)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-80161C	Bending Machines, Rotary Head, Tubing, Hydraulically Operated, Numerically Controlled

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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